Ground Before Correcting

Evidence-Guided Visual Chinese Grammatical Error Correction at Multi-LLL 2026

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The task requires text and spatial edits

Input: an image of a handwritten Chinese sentence

Output: corrected text + operation, replacement, and pixel box

Internal candidate: q = (recognized source, target, edits)

Released labels: INSERT, DELETE, SUBSTITUTE

Source: Sections 1 and 3.1, pp. 1–2

Why a single VLM pass is fragile

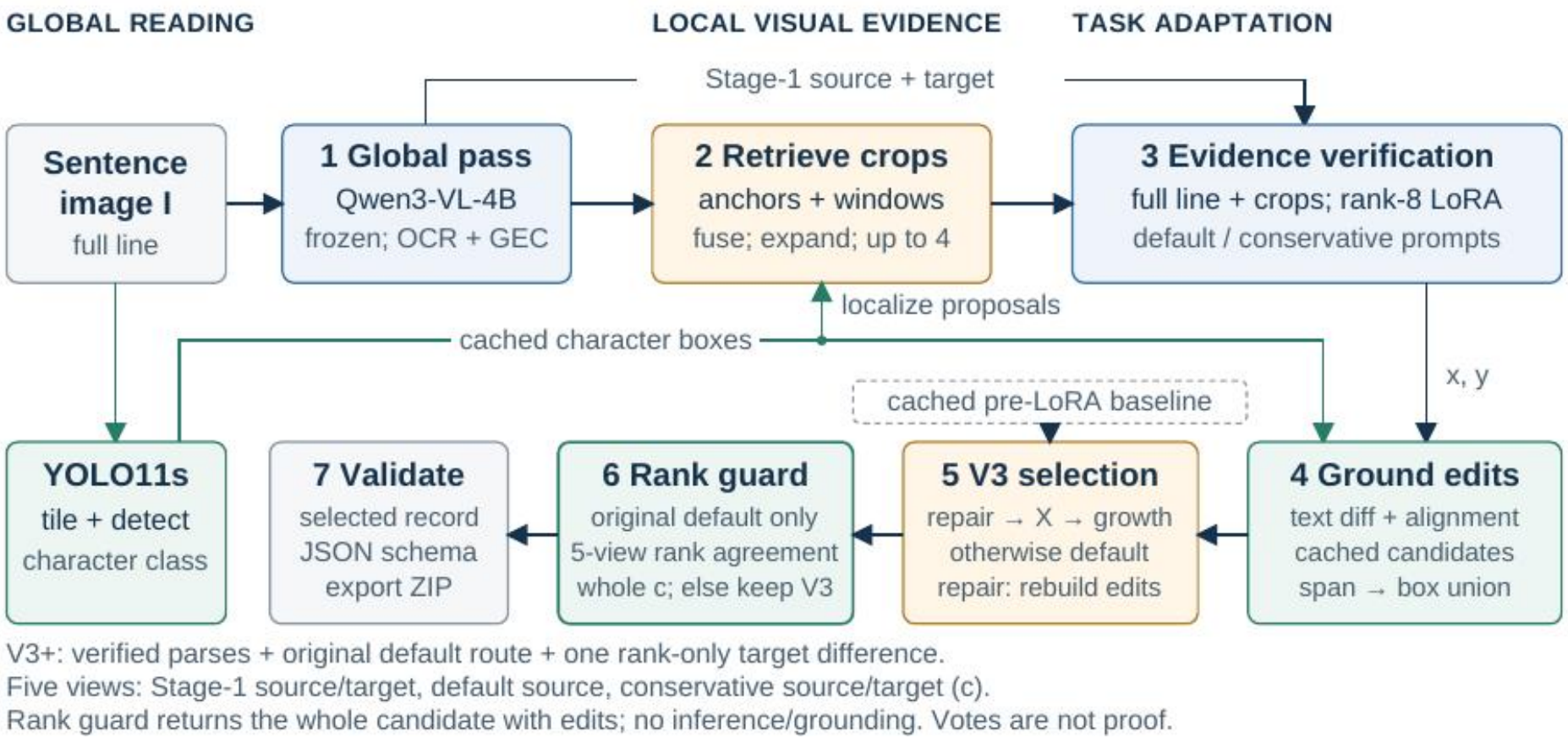
Long, narrow images lose character detail after resizing

Fluent corrections can contain unreliable coordinates

Extra rewriting can alter facts or hide necessary edits

Source: Introduction, p. 1; Section 4.7, p. 7

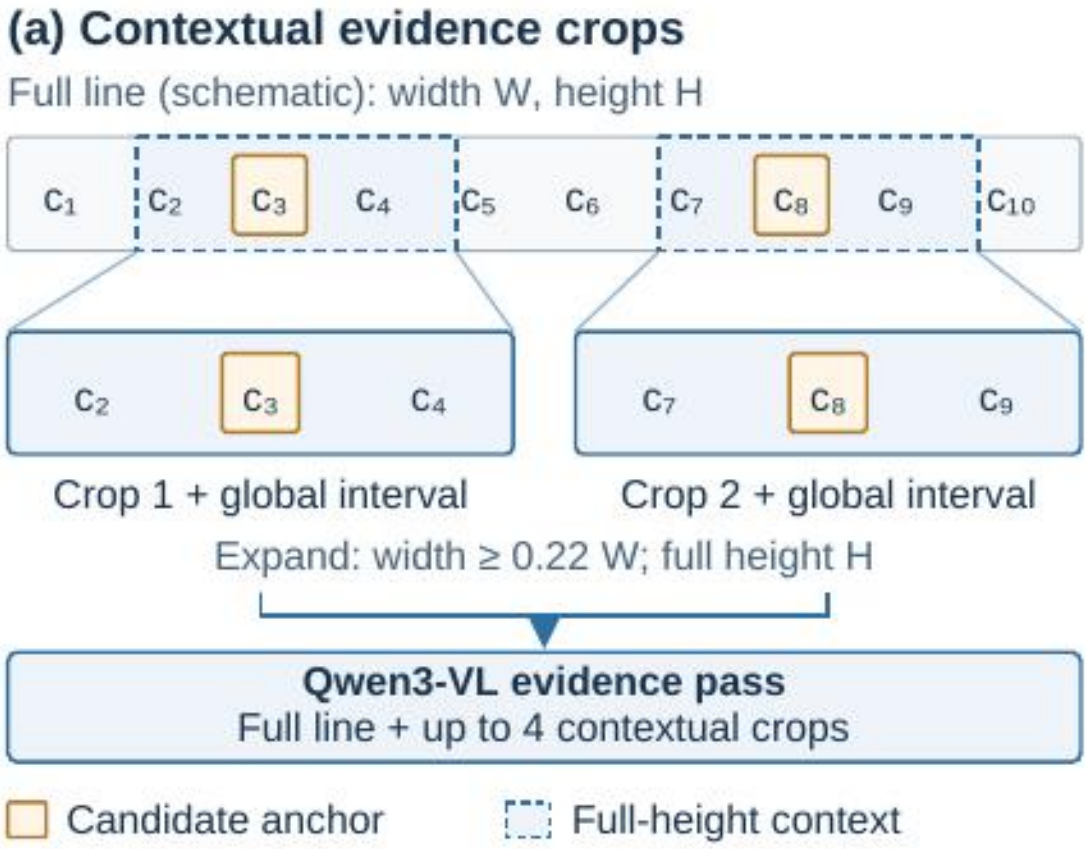
GroundCorrect pipeline



Rank guard: post-submission V3+ only

Source: Figure 1 and Section 3, pp. 2–4

Contextual crops preserve local evidence



Anchors: source–target differences and VLM proposals

Merge proposals and add overlapping coverage windows

At most 4 crops, full height, width ≥ 22% of the line

Test-time retrieval uses no gold target or gold edit boxes

Source: Section 3.2 and Figure 2(a), pp. 3–4

Format-matched LoRA verification

Input: full line, crops, coordinates, Stage-1 source and target

Training answer: source_text + target_text JSON

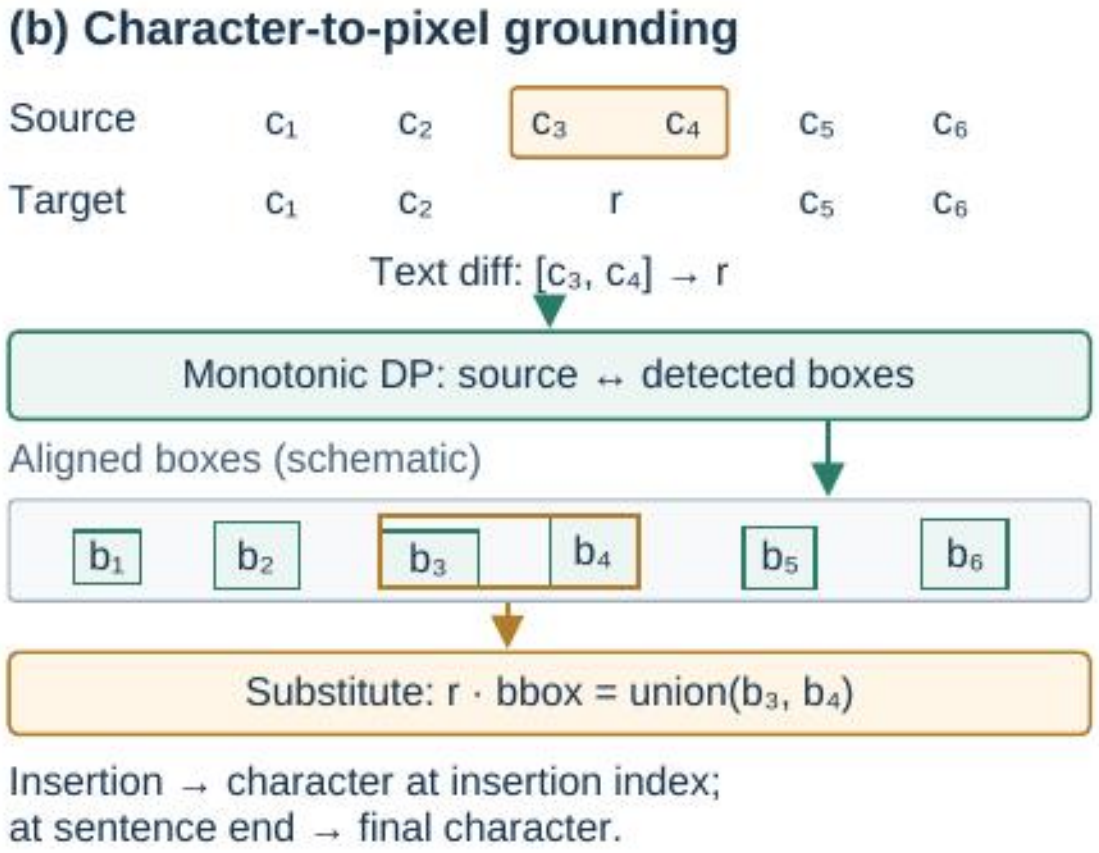
Rank 8, alpha 16, 16.52M trainable parameters (0.371%)

Assistant-only token loss; base model and vision encoder frozen

Invalid or empty output falls back to the Stage-1 hypothesis

Source: Section 3.2, p. 3; Appendix A.1, p. 10

Deterministic character-to-pixel grounding



Text differences determine edit spans

YOLO11s supplies boxes sorted from left to right

Monotonic DP aligns source characters to boxes

A non-insertion edit uses the union of available aligned boxes

Source: Section 3.3 and Figure 2(b), pp. 3–4

V3 finalization follows a fixed priority

1 **Narrow consistency repair: rebuild the edited record**

2 Newly introduced uppercase X: choose a safer candidate

3 Target growth ≥ 3 characters: choose a safer candidate

4 Otherwise retain the default candidate

Selection copies the complete source, target, and edit list

Source: Section 3.4, p. 4; Appendix B.1–B.3, pp. 11–12

Data use limits the evaluation scope

Subset	Images	LoRA fit	Detector fit
LoRA training	1,746	Yes	Yes
Internal selection	214	No	Yes
Development	245	No	Validation
Hidden test	246	No	No

233 / 245 dev sentences share essay groups with training

Neither local split is an untouched end-to-end test.

Source: Table 1 and Section 4.1, pp. 4–5

The local proxy scores text and grounded edits

Sproxy = mean over sentences of (text F0.5 + edit F0.5) / 2

Text match: operation, source span, replacement string

Grounded match: operation + replacement + box IoU ≥ 0.5

F0.5 emphasizes precision; both components are macro-averaged

The private official evaluator is unavailable

Source: Section 4.1 and Equation 6, p. 5

Controlled adapter gains under fixed inputs

Fixed configuration	Text F0.5	Edit F0.5	Composite
Dev: adapter OFF	0.2047	0.0460	0.1254
Dev: adapter ON	0.2770	0.1699	0.2234
Holdout: OFF	0.2034	0.0473	0.1253
Holdout: ON	0.2661	0.1585	0.2123

Dev gain +0.0981 95% interval [0.0621, 0.1337]

Same inputs, prompt, decoding, and grounding. No guard.
Gain uses unrounded scores. Intervals condition on one trained adapter.

Source: Table 3 and Section 4.3, p. 6

Official outcome and local progression

Official V3: 2nd place, score 0.5114 (retained team record)

Local reconstruction is not guaranteed byte-identical

Cached dev progression: 0.1641 → 0.2233 → 0.2243

That progression changes configurations, including the prompt

V3+ has no official hidden-test score

Source: Sections 4.2–4.4 and Table 2, p. 6

Correct text can still miss the required edit

(a) OCR recovery: improved text, incomplete localization 543_6

Input image 

		F0.5: text / edit
Reference	然后假期期间我喜欢在家里休息。	
Frozen S2	然后假期期间我改为在家里休息。	0.56 / 0.00
V3+	然后假期期间我喜欢在家里休息。	1.00 / 0.00

V3+ = V3: text matches the reference, but source normalization hides the gold edit.

(b) Missing-character insertion: correct text and localization 1969_7

Input image 

		F0.5: text / edit
Reference	手机和电脑不能消失的理由有很多。	
Frozen S2	手机和电脑不能消失的有很多。	0.00 / 0.00
V3+	手机和电脑不能消失的理由有很多。	1.00 / 1.00

V3+ = V3: the missing character is inserted; the baseline deletes its neighbor.

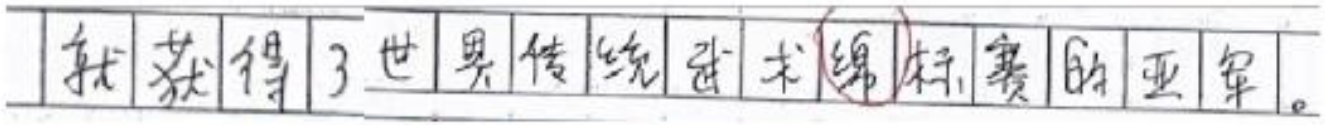
Faithful source recognition is necessary for complete grounding

Source: Figure 3(a,b) and Section 4.7, pp. 7–8

V3+ is a targeted rank-correction case study

(c) Rank protection: correct text, remaining localization failure 1587_15

Input image



		F0.5: text / edit
Reference	就获得了世界传统武术锦标赛的亚军。	
Old V3	就获得了世界传统武术锦标赛的冠军。	0.56 / 0.00
V3+	就获得了世界传统武术锦标赛的亚军。	1.00 / 0.00

V3+ restores runner-up via the whole conservative record; the gold edit remains unlocalized.

Only an original default route may switch to the whole conservative record

1 of 245 dev examples changes; 214 holdout records stay identical

Dev gain: +0.000907; grounded-edit score stays unchanged

Source: Sections 3.5, 4.5, 4.7; Figure 3(c), pp. 4, 7–8

Takeaways and open questions

Local evidence verification and separate grounding offer a clear task decomposition

The controlled comparison supports an adapter effect under the tested configuration

Recognition can hide edits even when the target is correct

Next tests: fully separated data, repeated seeds, and source-faithfulness checks

Source: Conclusion and Limitations, pp. 7–8